

same direction. Since spring 2007 correlations have been on the rise and only recently seem to have leveled off, but at much higher levels than at the beginning of the decade. An interesting finding of our historical analysis is that the market experienced the highest level of correlation following the flash crash (but very short lived) and again during the US debt ceiling crisis. These correlation levels were even higher than the levels experienced during the financial crisis when markets experienced their highest levels of volatility.

The average large cap pair-wise correlation during our sample period was 32% and average small cap pair-wise correlation was 23%. Opposite to the volatility relationship mentioned above, small cap correlation tends to be lower than large cap correlation. This is primarily due to smaller companies having a larger amount of company risk. However, there are periods when correlation is greater across the small cap stocks.

Dispersion: We measured dispersion as the standard deviation of the 22-day stock return. For example, for the SP500 we computed the 22-day price return for each of the 500 stocks in the index. We then computed the standard deviation of these returns as our dispersion. [Figure 6.2c](#) illustrates large cap returns dispersion over our analysis period. As shown in the figure, dispersion was fairly constant at the beginning of the decade and then declined through August 2007 when it suddenly increased and spiked during the financial crisis. Dispersion decreased shortly thereafter but began increasing again during the US debt crisis of fall 2011. Comparison of dispersion to pair-wise correlation shows a slightly negative relationship. As correlation increases dispersion tends to fall and as correlation decreases dispersion tends to increase. This inverse relationship between dispersion and correlation is more dramatic for small cap stocks than for large cap stocks.

Over our sample period, the average large cap dispersion was 36% and the average small cap dispersion was 58%. Dispersion tends to be consistently larger for small caps than for large caps.

Volatility Clustering: [Figure 6.2d](#) illustrates the daily price change for the SP500 index. Notice the extent of volatility clustering. This means that large price swings tend to be followed by large price swings (either positive or negative) and small price swings tend to be followed by small price swings (either positive or negative). The beginning of the decade was associated with a higher volatility regime due to the technology bubble. This was followed by relatively lower volatility through the beginning of the quant breakdown starting in August 2007 followed by ultra-high volatility during the financial crisis of 2008–2009. The graph also